

LAB 8: Configuring DHCP, DNS, and Web Services on a Server Using Cisco Packet Tracer

OBJECTIVES:

- To set up a server that provides DHCP, DNS, and HTTP (web) services within Cisco Packet Tracer.
- To learn how DHCP automatically assigns IP addresses and how DNS resolves domain names to IPs.
- To confirm that server services are functioning correctly by accessing a web page from connected client devices.

THEORY:

1. Server Setup and Role:

A server is a dedicated system that provides network services to client devices. For reliability, servers are usually given static IP addresses. In this exercise, a single server is configured to offer DHCP, DNS, and web (HTTP) services. Proper server setup allows centralized management, smooth communication, and easier administration of the network.

2. Dynamic Host Configuration Protocol (DHCP):

DHCP is a protocol that automatically provides IP addresses and other network settings—like subnet mask, default gateway, and DNS server—to client devices. This automation removes the need for manual IP setup, reduces errors, and supports network scalability. In this lab, the server dynamically assigns IP addresses to the connected PCs.

3. Domain Name System (DNS):

DNS converts human-friendly domain names into IP addresses. Instead of remembering numbers like 192.168.1.1, users can use names like www.hcoe.com. The DNS server stores these mappings and responds to clients' queries to direct them to the correct IP address.

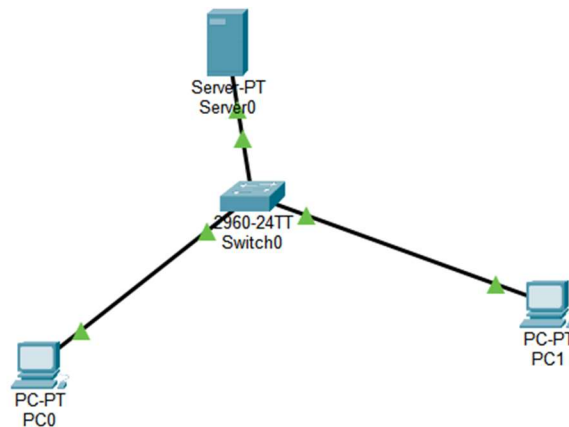
4. Web Server (HTTP Service):

A web server hosts web pages and provides them to clients via the HTTP protocol. It handles browser requests and serves content such as HTML files. In this lab, enabling the HTTP service allows client PCs to access a webpage through a browser.

5. Combining DHCP, DNS, and Web Services:

These services complement each other for efficient network operation. DHCP handles automatic IP assignment, DNS resolves domain names to IP addresses, and the web server delivers web content. Their integration illustrates how centralized services simplify network setup and enhance usability.

NETWORK TOPOLOGY:



IP CONFIGURATION:

IP addressing scheme

Device	IP Address	Subnet Mask	Gateway
Server	192.168.1.2	255.255.255.0	192.168.1.1
PC	DHCP	DHCP	DHCP

For PCs

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC1	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

For Server

Device	Default Gateway	DNS Server
Server0	192.168.1.1	192.168.1.2

SERVER CONFIGURATION:

Step 1: Assign a Static IP to the Server

1. Select the **Server** device in Cisco Packet Tracer.
2. Navigate to **Desktop → IP Configuration**.
3. Enter the following network details:
 - **IP Address:** 192.168.1.2
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.1.1
 - **DNS Server:** 192.168.1.2

Step 2: Activate DHCP Service

1. On the server, go to **Services → DHCP**.
2. Switch the **DHCP service** to **ON**.

Step 3: Configure a DHCP Pool

Fill in the DHCP pool information as follows and click **Add**:

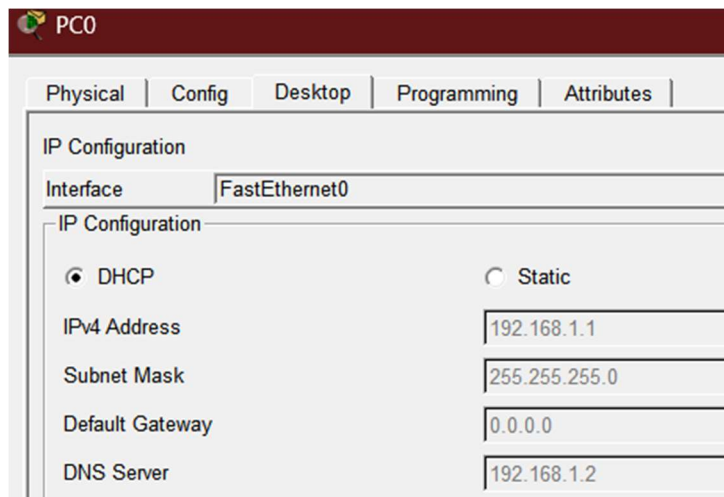
Field	Value
Pool Name	LAB_POOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Starting IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

Click Add

Verification:

On each PC:

- Go to Desktop → IP Configuration
- Select DHCP



The screenshot shows the configuration window for PC0. The 'Desktop' tab is selected, and the 'IP Configuration' section is expanded. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'DHCP' radio button is selected, and the 'Static' radio button is unselected. The following fields are filled in:

IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	192.168.1.2

Step 4: Activate DNS Service

1. Select the **Server** device.
2. Navigate to **Services → DNS**.
3. Set the DNS service to **ON**.

Step 5: Create a DNS Record

Enter the following information to map the domain name to the server's IP address:

- Name: www.hcoe.com
- Record Type: A
- IP Address: 192.168.1.2

Click Add to save the record.

DNS Verification

On each PC:

- Open **Command Prompt**
- Type: ping www.hcoe.com

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Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.hcoe.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Web Server (HTTP) Configuration

Step 6: Enable HTTP Service

1. Go to **Server → Services → HTTP**.
2. Switch the HTTP service **ON**.
3. (Optional) Modify the **index.html** file to customize the webpage content.

Web Server Verification

On a client PC:

- Open **Desktop → Web Browser**
- Enter the URL: www.hcoe.com

If everything is configured correctly, the webpage hosted on the server will load successfully.



RESULT:

The server was properly configured to provide DHCP, DNS, and HTTP services. All connected client PCs obtained IP addresses automatically from the DHCP pool without manual configuration. The DNS service successfully translated the domain name into the corresponding IP address, and the hosted webpage was accessed through a browser without any errors. This confirms that all configured services operated correctly.

DISCUSSION:

This experiment provided hands-on experience in setting up core network services within a simulated environment. DHCP reduced manual configuration by dynamically assigning network parameters. DNS allowed devices to communicate using easy-to-remember domain names instead of numerical IP addresses. The HTTP service demonstrated application-layer functionality by delivering web content to clients. The lab also highlighted how these services depend on one another to ensure smooth network communication and efficient centralized management.

CONCLUSION:

The lab objectives were successfully accomplished by configuring and verifying DHCP, DNS, and web services in Cisco Packet Tracer. The experiment confirmed accurate IP allocation, proper domain name resolution, and successful web access from client systems. Overall, the activity reinforced the significance of integrated server-based services in modern networking environments.